

Cell Design Specification - Long-Endurance Drone Battery

VBF-T8R1NOFORCE-DS-01 | case t8_r1_noforce | 2026-08-25 | virtual design (simulation-based)

1. Basic specification

Item	Value	Source
Electrochemical system	NMC811 / graphite (Chen2020 parameterization)	parameter set (base selection per anchor table)
Nominal capacity	5.0 Ah (set) / 5.2725 Ah (simulated 1C)	r6_v9_1c_dfn.json capacity_ah
Voltage window	2.5 - 4.2 V	parameter set
Electrode dimensions	65 mm height x 1891 mm width x 0.1848 mm stack	parameter set + r6_v9_energy.json
Electrolyte	1 M LiPF6 EC:DMC 1:1; kappa 1.4 S/m, D 3.7e-10 m2/s (literature, Nyman 2008); t+ 0.2594 (set)	r6_v9_kin_el.json + set
Cooling requirement	h = 60 W/m2/K forced-air (ducted prop-wash)	r6_v9_kin_el.json
Shell/casing thickness	not provided (no parameter)	-

2. Electrode and separator

Layer	Thickness (um)	Porosity	AM vol. frac.	Particle radius (um)
Positive (NMC811)	75.6	0.40	0.60	0.7 (nano-NMC)
Negative (graphite)	85.2	0.35	0.65	1.2 (fine graphite)
Separator	10	0.47	-	-
Positive current collector (Al)	8	-	-	-
Negative current collector (Cu)	6	-	-	-

$N/P = (33133 \times 0.65 \times 85.2e-6) / (63104 \times 0.60 \times 75.6e-6) = 0.641$ (capacity-density basis, mechanical). The cell is negative-limited - a deliberate transport trade (see design notes).

3. Process design parameters

Parameter	Formula	Value
Positive areal density	$t \times (1-e) \times \rho$	$75.6e-6 \times 0.60 \times 3262 = 147.96 \text{ g/m}^2$
Negative areal density	$t \times (1-e) \times \rho$	$85.2e-6 \times 0.65 \times 1657 = 91.76 \text{ g/m}^2$
Positive compaction density	$\rho \times (1-e)$	$3262 \times 0.60 = 1.96 \text{ g/cm}^3$
Negative compaction density	$\rho \times (1-e)$	$1657 \times 0.65 = 1.08 \text{ g/cm}^3$
Electrolyte fill amount	pore vol x density x fill	$7.960e-6 \text{ m}^3 \times 1200 \text{ kg/m}^3 \times 1.0 = 9.55 \text{ g}$ (density literature)
Formation (recommended)	-	0.1C CC to 4.2 V, 25 C, 2 cycles (design rec.; production tuning required)

4. Mass breakdown (contract caliber; electrolyte excluded per calc-energy)

Layer	kg/m2	g/cell (area 0.122915 m2)
Positive electrode (active)	0.14796	18.19
Negative electrode (active)	0.09176	11.28
Positive current collector (Al 8 um)	0.02160	2.65
Negative current collector (Cu 6 um)	0.05376	6.61
Separator (10 um)	0.00210	0.26
TOTAL (contract)	0.31718	38.99 g (criterion: <= 40 g PASS)
Electrolyte (excluded)	-	9.55 g (BOM row, literature density)

5. Performance verification (entry-0 criteria)

Metric	Result	Criterion	Verdict
Energy density	497.97 Wh/kg	≥ 446.18 Wh/kg	PASS
5C capacity retention	0.9896	≥ 0.90	PASS
Cell mass	38.99 g	≤ 40 g	PASS
Lithium plating (4C charge, 45 C)	anode min +0.00635 V	plated = false	PASS (margin +6.3 mV)
T_max 4C charge	326.81 K	≤ 333.15 K	PASS
T_max 5C discharge	313.64 K	≤ 333.15 K	PASS
1C capacity	5.2725 Ah	no threshold (feeds ED)	informational

6. Design notes (change log, cited from log.jsonl)

* R1 baseline (Chen2020 defaults): ED 400.75, 5C retention 0.087, mass 43.45 g - all three fail; diagnosis: solid-diffusion-limited at 5C ($\tau \sim r^2$) + inert mass 0.153 kg/m².

* R2 attribution: inert lean-out solves ED/mass but not rate; particles 0.2325; porosity 0.1998.

* R3: rate levers combined (r_p 0.7 μ m, r_n 2 μ m, porosity 0.40/0.35, high-kappa electrolyte) -> retention 0.9524, ED 493.6, mass 33.0 g.

* R4: sizing (width 1.891 m -> 39.0 g) + thermal ($h=40$): stage2 pass, stage3 fail - plated (anode -0.030 V), 4C T_max 334.14 K.

* R5 (V8 thicker anode, N/P 1.13): backfired - anode min -0.194 V (plating dip is charge-end electrolyte transport polarization, not capacity) + 5C IDA_CONV_FAIL recorded verbatim.

* R6 (final V9): anode kinetics r_n 1.2 μ m + EC:DMC electrolyte (κ 1.4, D 3.7e-10) + $h=60$ -> anode min +0.00635 V, 4C T_max 326.81 K, retention 0.9896, ED 497.97, mass 38.99 g. All five criteria PASS.